

The Fibonacci Numbers

COMS10018 - Algorithms

Dr Christian Konrad

The Fibonacci Numbers

Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2 .$$

The Fibonacci Numbers

Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2 .$$

0 1 1 2 3 5 8 13 21 34 55 89 ...

The Fibonacci Numbers

Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2 .$$

0 1 1 2 3 5 8 13 21 34 55 89 ...

Why are they important?

The Fibonacci Numbers

Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2 .$$

0 1 1 2 3 5 8 13 21 34 55 89 ...

Why are they important?

- Fibonacci heaps (data structure)

The Fibonacci Numbers

Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2 .$$

0 1 1 2 3 5 8 13 21 34 55 89 ...

Why are they important?

- Fibonacci heaps (data structure)
- Appear in analysis of algorithms (e.g. Euclid's algorithm)

The Fibonacci Numbers

Fibonacci Numbers

$$F_0 = 0$$

$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2 .$$

0 1 1 2 3 5 8 13 21 34 55 89 ...

Why are they important?

- Fibonacci heaps (data structure)
- Appear in analysis of algorithms (e.g. Euclid's algorithm)
- Appear everywhere in nature

The Fibonacci Numbers

Fibonacci Numbers

$$F_0 = 0$$

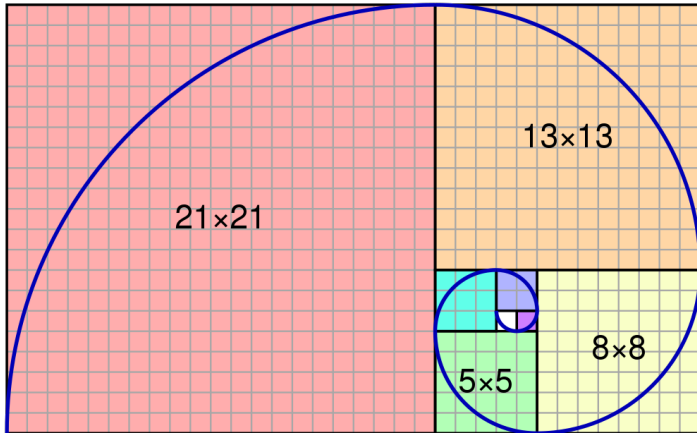
$$F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2.$$

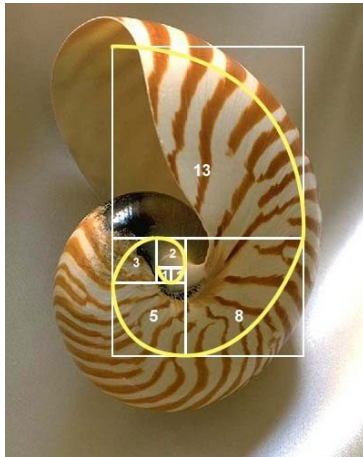
0 1 1 2 3 5 8 13 21 34 55 89 ...

Why are they important?

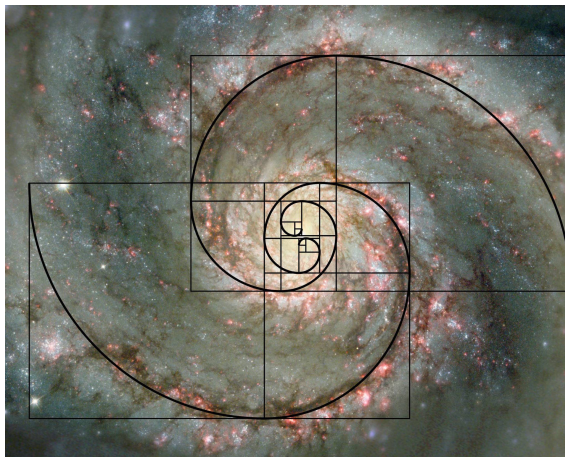
- Fibonacci heaps (data structure)
- Appear in analysis of algorithms (e.g. Euclid's algorithm)
- Appear everywhere in nature
- Interesting and instructive computational problem



source: wikipedia



source: [realworldmathematics](#) at [wordpress](#)



source: brian koberlein

Computing the Fibonacci Numbers

Naïve Algorithm

Require: Integer $n \geq 0$

if $n \leq 1$ **then**

return n

else

return $\text{FIB}(n-1) + \text{FIB}(n-2)$

$\text{FIB}(n)$

Computing the Fibonacci Numbers

Naïve Algorithm

```
Require: Integer  $n \geq 0$   
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n - 1) + \text{FIB}(n - 2)$ 
```

$\text{FIB}(n)$

What is the runtime of this algorithm?

Computing the Fibonacci Numbers

Naïve Algorithm

```
Require: Integer  $n \geq 0$   
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n - 1) + \text{FIB}(n - 2)$ 
```

$\text{FIB}(n)$

What is the runtime of this algorithm?

Runtime:

Computing the Fibonacci Numbers

Naïve Algorithm

```
Require: Integer  $n \geq 0$   
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n-1) + \text{FIB}(n-2)$ 
```

$\text{FIB}(n)$

What is the runtime of this algorithm?

Runtime:

- Without recursive calls, runtime is $O(1)$

Computing the Fibonacci Numbers

Naïve Algorithm

```
Require: Integer  $n \geq 0$   
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n-1) + \text{FIB}(n-2)$ 
```

$\text{FIB}(n)$

What is the runtime of this algorithm?

Runtime:

- Without recursive calls, runtime is $O(1)$
- Hence, runtime is $O(\text{"number of recursive calls"})$

Runtime Analysis

Define Recurrence:

$T(n)$: number of recursive calls to FIB when called with parameter n

```
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n-1) + \text{FIB}(n-2)$ 
```

Define Recurrence:

$T(n)$: number of recursive calls to FIB when called with parameter n

```
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n-1) + \text{FIB}(n-2)$ 
```

$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

Define Recurrence:

$T(n)$: number of recursive calls to FIB when called with parameter n

```
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n-1) + \text{FIB}(n-2)$ 
```

$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

How to Solve this Recurrence?

Define Recurrence:

$T(n)$: number of recursive calls to FIB when called with parameter n

```
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n-1) + \text{FIB}(n-2)$ 
```

$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

How to Solve this Recurrence?

- We will use the recursion tree technique to obtain a guess for an upper bound

Define Recurrence:

$T(n)$: number of recursive calls to FIB when called with parameter n

```
if  $n \leq 1$  then  
    return  $n$   
else  
    return  $\text{FIB}(n-1) + \text{FIB}(n-2)$ 
```

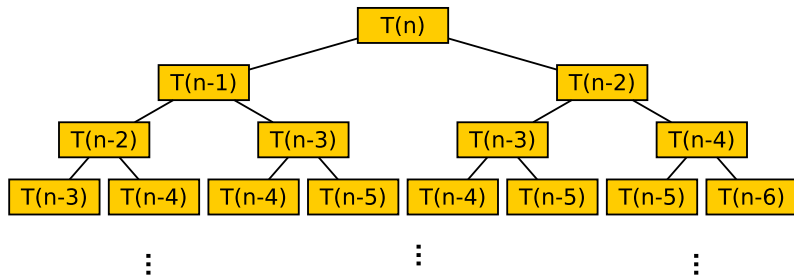
$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

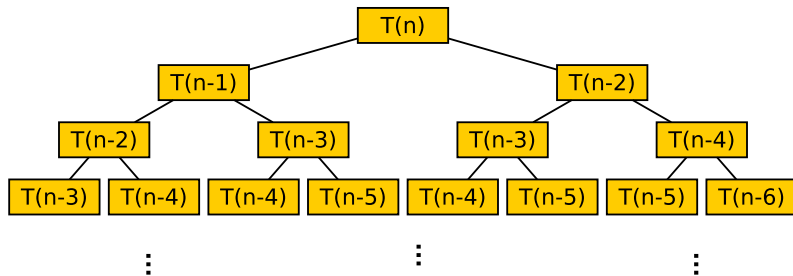
How to Solve this Recurrence?

- We will use the recursion tree technique to obtain a guess for an upper bound
- We will verify the guess with the substitution method

Recursion Tree for T



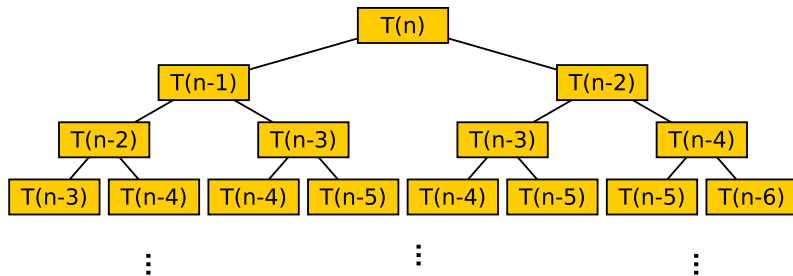
Recursion Tree for T



Observe:

- Each node contributes 1
- Hence, $T(n)$ equals number of nodes
- Number of levels of recursion tree:

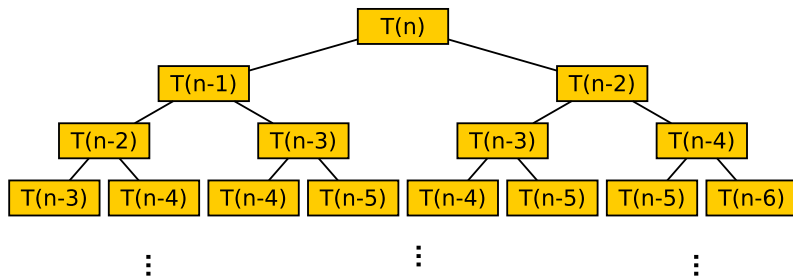
Recursion Tree for T



Observe:

- Each node contributes 1
- Hence, $T(n)$ equals number of nodes
- Number of levels of recursion tree: n
- Our guess:

Recursion Tree for T



Observe:

- Each node contributes 1
- Hence, $T(n)$ equals number of nodes
- Number of levels of recursion tree: n
- Our guess: $T(n) \leq c^n$ (we believe $c \leq 2$)

Verification with the Substitution Method

Recall:

$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

Our guess: $T(n) \leq c^n$

Substitute Guess into Recurrence:

Verification with the Substitution Method

Recall:

$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

Our guess: $T(n) \leq c^n$

Substitute Guess into Recurrence:

$$T(n) = 1 + T(n-1) + T(n-2) \leq$$

Verification with the Substitution Method

Recall:

$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

Our guess: $T(n) \leq c^n$

Substitute Guess into Recurrence:

$$T(n) = 1 + T(n-1) + T(n-2) \leq 1 + c^{n-1} + c^{n-2}$$

- It is required that $1 + c^{n-1} + c^{n-2} \leq c^n$

Verification with the Substitution Method

Recall:

$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

Our guess: $T(n) \leq c^n$

Substitute Guess into Recurrence:

$$T(n) = 1 + T(n-1) + T(n-2) \leq 1 + c^{n-1} + c^{n-2}$$

- It is required that $1 + c^{n-1} + c^{n-2} \leq c^n$
- The additive 1 prevents us from getting a similar form as c^n

Verification with the Substitution Method

Recall:

$$T(0) = T(1) = 1$$

$$T(n) = 1 + T(n-1) + T(n-2), \text{ for } n \geq 2.$$

Our guess: $T(n) \leq c^n$

Substitute Guess into Recurrence:

$$T(n) = 1 + T(n-1) + T(n-2) \leq 1 + c^{n-1} + c^{n-2}$$

- It is required that $1 + c^{n-1} + c^{n-2} \leq c^n$
- The additive 1 prevents us from getting a similar form as c^n
- Try different guess: $T(n) \leq c^n - 1$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$T(n) = 1 + T(n-1) + T(n-2)$$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1 . \end{aligned}$$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1 . \end{aligned}$$

Select smallest possible c :

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1. \end{aligned}$$

Select smallest possible c :

$$c^{n-1} + c^{n-2} = c^n$$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1. \end{aligned}$$

Select smallest possible c :

$$\begin{aligned} c^{n-1} + c^{n-2} &= c^n \\ 0 &= c^2 - c - 1 \end{aligned}$$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1 . \end{aligned}$$

Select smallest possible c :

$$\begin{aligned} c^{n-1} + c^{n-2} &= c^n \\ 0 &= c^2 - c - 1 \\ c &= \frac{1 + \sqrt{5}}{2} \approx 1.618033989 . \end{aligned}$$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1 . \end{aligned}$$

Select smallest possible c :

$$\begin{aligned} c^{n-1} + c^{n-2} &= c^n \\ 0 &= c^2 - c - 1 \\ c &= \frac{1 + \sqrt{5}}{2} \approx 1.618033989 . \text{ Golden Ratio!} \end{aligned}$$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1. \end{aligned}$$

Select smallest possible c :

$$\begin{aligned} c^{n-1} + c^{n-2} &= c^n \\ 0 &= c^2 - c - 1 \\ c &= \frac{1 + \sqrt{5}}{2} \approx 1.618033989. \text{ Golden Ratio!} \end{aligned}$$

Base Case:

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1. \end{aligned}$$

Select smallest possible c :

$$\begin{aligned} c^{n-1} + c^{n-2} &= c^n \\ 0 &= c^2 - c - 1 \\ c &= \frac{1 + \sqrt{5}}{2} \approx 1.618033989. \text{ Golden Ratio!} \end{aligned}$$

Base Case:

- $T(0) = T(1) = 1$

Verification with the Substitution Method (2)

New Guess: $T(n) \leq c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (c^{n-1} - 1) + (c^{n-2} - 1) = c^{n-1} + c^{n-2} - 1. \end{aligned}$$

Select smallest possible c :

$$\begin{aligned} c^{n-1} + c^{n-2} &= c^n \\ 0 &= c^2 - c - 1 \\ c &= \frac{1 + \sqrt{5}}{2} \approx 1.618033989. \text{ Golden Ratio!} \end{aligned}$$

Base Case:

- $T(0) = T(1) = 1$
- $c^0 - 1 = 0$ and $c^1 - 1 \approx 0.61$ ✗

Verification with the Substitution Method (3)

Another New Guess:

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

$$T(n) = 1 + T(n-1) + T(n-2)$$

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (k \cdot c^{n-1} - 1) + (k \cdot c^{n-2} - 1) \end{aligned}$$

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

$$\begin{aligned} T(n) &= 1 + T(n-1) + T(n-2) \\ &\leq 1 + (k \cdot c^{n-1} - 1) + (k \cdot c^{n-2} - 1) \\ &= k(c^{n-1} + c^{n-2}) - 1. \end{aligned}$$

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

$$\begin{aligned}T(n) &= 1 + T(n-1) + T(n-2) \\&\leq 1 + (k \cdot c^{n-1} - 1) + (k \cdot c^{n-2} - 1) \\&= k(c^{n-1} + c^{n-2}) - 1.\end{aligned}$$

Select smallest possible c : $c = \frac{1+\sqrt{5}}{2}$ as before

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

$$\begin{aligned}T(n) &= 1 + T(n-1) + T(n-2) \\&\leq 1 + (k \cdot c^{n-1} - 1) + (k \cdot c^{n-2} - 1) \\&= k(c^{n-1} + c^{n-2}) - 1.\end{aligned}$$

Select smallest possible c : $c = \frac{1+\sqrt{5}}{2}$ as before

Base Case:

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

$$\begin{aligned}T(n) &= 1 + T(n-1) + T(n-2) \\&\leq 1 + (k \cdot c^{n-1} - 1) + (k \cdot c^{n-2} - 1) \\&= k(c^{n-1} + c^{n-2}) - 1.\end{aligned}$$

Select smallest possible c : $c = \frac{1+\sqrt{5}}{2}$ as before

Base Case:

- $T(0) = T(1) = 1$
- $k \cdot c^0 - 1 = k - 1$ and $k \cdot c^1 - 1 > k - 1 \checkmark$

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

$$\begin{aligned}T(n) &= 1 + T(n-1) + T(n-2) \\&\leq 1 + (k \cdot c^{n-1} - 1) + (k \cdot c^{n-2} - 1) \\&= k(c^{n-1} + c^{n-2}) - 1.\end{aligned}$$

Select smallest possible c : $c = \frac{1+\sqrt{5}}{2}$ as before

Base Case:

- $T(0) = T(1) = 1$
- $k \cdot c^0 - 1 = k - 1$ and $k \cdot c^1 - 1 > k - 1 \checkmark$
- We can hence select $k = 2$!

Verification with the Substitution Method (3)

Another New Guess: $T(n) \leq k \cdot c^n - 1$

$$\begin{aligned}T(n) &= 1 + T(n-1) + T(n-2) \\&\leq 1 + (k \cdot c^{n-1} - 1) + (k \cdot c^{n-2} - 1) \\&= k(c^{n-1} + c^{n-2}) - 1.\end{aligned}$$

Select smallest possible c : $c = \frac{1+\sqrt{5}}{2}$ as before

Base Case:

- $T(0) = T(1) = 1$
- $k \cdot c^0 - 1 = k - 1$ and $k \cdot c^1 - 1 > k - 1 \checkmark$
- We can hence select $k = 2$!

We proved $T(n) \leq 2 \cdot \left(\frac{1+\sqrt{5}}{2}\right)^n - 1$. Hence $T(n) \in O\left(\left(\frac{1+\sqrt{5}}{2}\right)^n\right)$.

Fibonacci Numbers: Closed-form Expression

Golden Ratio:

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.61803$$

Fibonacci Numbers: Closed-form Expression

Golden Ratio:

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.61803$$

Closed-form Expression:

$$F_n = \frac{\phi^n - (-\phi)^{-n}}{\sqrt{5}} .$$

Fibonacci Numbers: Closed-form Expression

Golden Ratio:

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.61803$$

Closed-form Expression:

$$F_n = \frac{\phi^n - (-\phi)^{-n}}{\sqrt{5}} .$$

Why not compute Fibonacci Numbers this way?

Fibonacci Numbers: Closed-form Expression

Golden Ratio:

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.61803$$

Closed-form Expression:

$$F_n = \frac{\phi^n - (-\phi)^{-n}}{\sqrt{5}} .$$

Why not compute Fibonacci Numbers this way?

- Floating point operations, precision

Fibonacci Numbers: Closed-form Expression

Golden Ratio:

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.61803$$

Closed-form Expression:

$$F_n = \frac{\phi^n - (-\phi)^{-n}}{\sqrt{5}} .$$

Why not compute Fibonacci Numbers this way?

- Floating point operations, precision
- Large numbers involved

Fibonacci Numbers: Closed-form Expression

Golden Ratio:

$$\phi = \frac{1 + \sqrt{5}}{2} \approx 1.61803$$

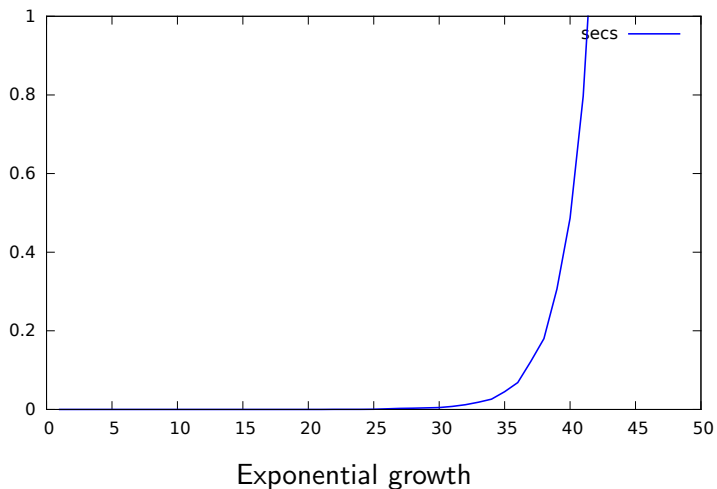
Closed-form Expression:

$$F_n = \frac{\phi^n - (-\phi)^{-n}}{\sqrt{5}} .$$

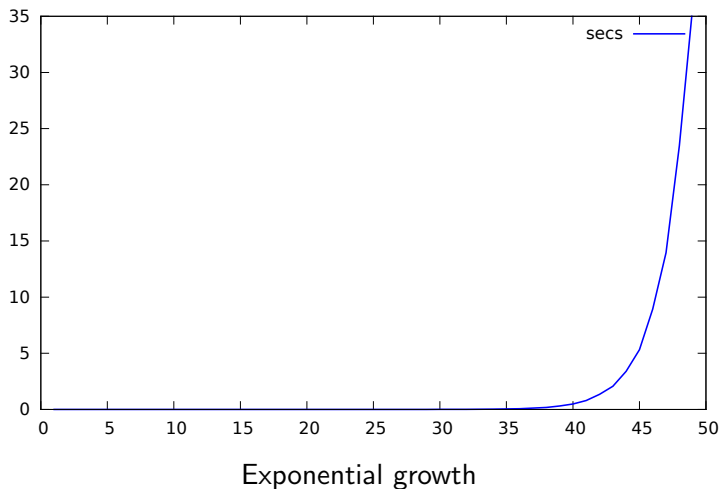
Why not compute Fibonacci Numbers this way?

- Floating point operations, precision
- Large numbers involved
- Impractical

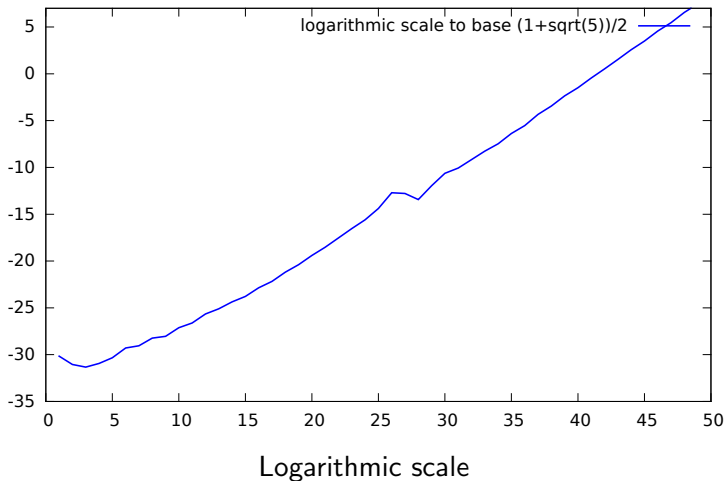
Experiments



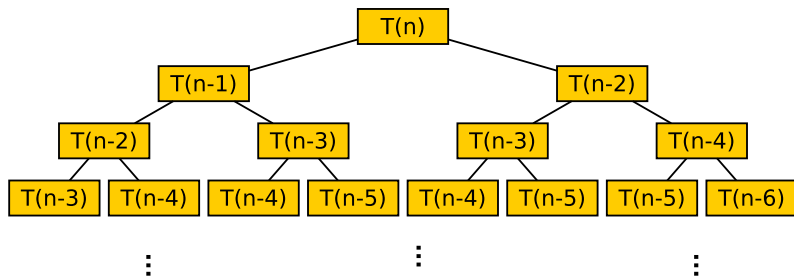
Experiments



Experiments

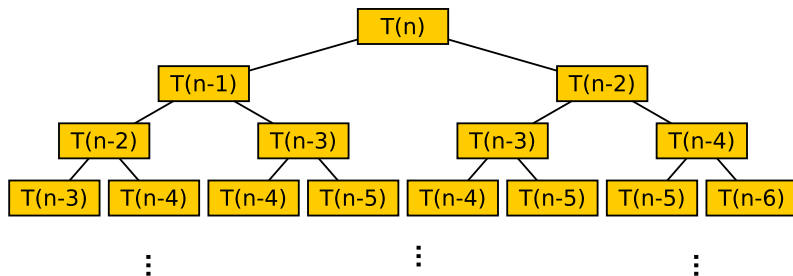


Why is this Algorithm so slow?



Discussion:

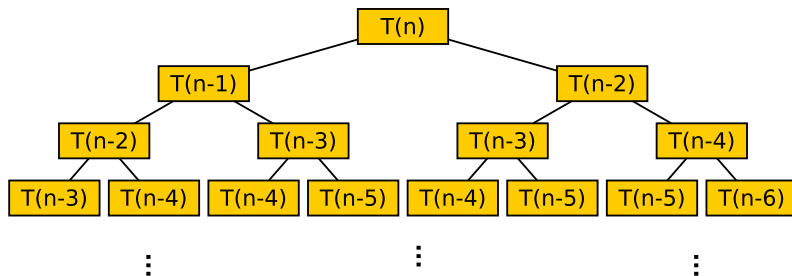
Why is this Algorithm so slow?



Discussion:

- We compute solutions to subproblems many times ($T(i)$ is computed often, for most values of i)

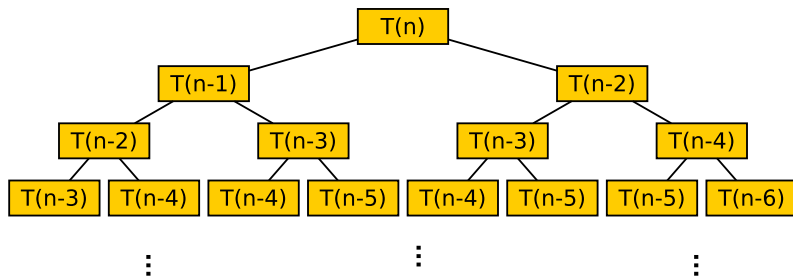
Why is this Algorithm so slow?



Discussion:

- We compute solutions to subproblems many times ($T(i)$ is computed often, for most values of i)
- How can we avoid this?

Why is this Algorithm so slow?



Discussion:

- We compute solutions to subproblems many times ($T(i)$ is computed often, for most values of i)
- How can we avoid this?

Dynamic Programming!

Dynamic Programming Solution

Dynamic Programming (will be discussed in more detail later)

- Store solutions to subproblems in a table
- Compute table bottom up

Require: Integer $n \geq 0$

if $n \leq 1$ **then**

return n

else

$A \leftarrow$ array of size n

$A[0] \leftarrow 0, A[1] \leftarrow 1$

for $i \leftarrow 2 \dots n$ **do**

$A[i] \leftarrow A[i - 2] + A[i - 1]$

return $A[n]$

DYNPRGFIB(n)

Analysis:

- DynPrgFib() runs in time $O(n)$
- It uses space $\Theta(n)$ since it uses an array of size n

Analysis:

- DynPrgFib() runs in time $O(n)$
- It uses space $\Theta(n)$ since it uses an array of size n

Can we reduce the space to $O(1)$?

Analysis:

- DynPrgFib() runs in time $O(n)$
- It uses space $\Theta(n)$ since it uses an array of size n

Can we reduce the space to $O(1)$?

Improvement:

- Observe that when $T(i)$ is computed, the values $T(1), T(2), \dots, T(i-3)$ are no longer needed
- Only store the last two values of T

Improved Algorithm

```
Require: Integer  $n \geq 0$   
if  $n \leq 1$  then  
    return  $n$   
else  
     $a \leftarrow 0$   
     $b \leftarrow 1$   
    for  $i \leftarrow 2 \dots n$  do  
         $c \leftarrow a + b$   
         $a \leftarrow b$   
         $b \leftarrow c$   
    return  $c$ 
```

IMPROVEDDYNPRGFIB(n)

Correctness: via loop invariant!